

# SHAHADAT HUSSAIN, PhD

## MATERIALS ENGINEER – CHARACTERIZATION & METALLURGY

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### PROFESSIONAL SUMMARY

Materials Engineer with a PhD in Materials Science and 6+ years of post-doctoral and applied R&D experience leading materials characterization, metallography, failure analysis, and mechanical testing programs. Hands-on technical operator of advanced characterization suites—including Scanning Electron Microscopy (JEOL, FIB-SEM with EDS), X-ray Diffraction (Bruker, Rigaku), and Differential Scanning Calorimetry—alongside static and dynamic load frames (Instron, MTS). Proven track record in physical metallurgy, alloy phase transformations, microstructural defect mapping, and fatigue qualification of metallic components. Combines metallurgical depth with Python/MATLAB data science skills to establish reproducible laboratory methods, support quality systems, and optimize material performance. Seeking to bring this characterization expertise and laboratory ownership to Ovako Group’s R&D organization.

### CORE COMPETENCIES

- **Materials Characterization & Microscopy:** SEM, FIB-SEM (SEI/BSE/EDS), XRD (Bruker D2, Rigaku MiniFlex), DSC, Optical Microscopy, Quantitative Image Analysis
- **Metallography & Sample Preparation:** Precision Sectioning, Mounting, Mechanical Polishing, Chemical Etching (HF/Nitric Acid), Grain Size & Phase Fraction Analysis
- **Mechanical & Fatigue Testing:** Static Tension/Compression (Instron 5969, MTS 100 kN), High-Cycle Fatigue Qualification (Instron 8872), Hardness Testing (Vickers/Rockwell)
- **Physical Metallurgy & Processing:** Phase Transformation Behavior, Microstructural Inhomogeneity, Heat Treatment (Quenching, Annealing), Rolling, Vacuum Induction Melting
- **Data Science & Method Validation:** Python (Pandas, SciPy, Scikit-learn), MATLAB, Test Method Standardization, Statistical Analysis
- **Laboratory Operations & EHS:** Equipment Maintenance and Calibration, Chemical Safety, HF Handling Protocols, Risk Assessment
- **Manufacturing Processes:** Additive Manufacturing, Hybrid Manufacturing, Investment Casting, Induction Melting Die Casting

### PROFESSIONAL EXPERIENCE

#### Associate Research Scientist — Advanced Materials & Characterization

Sep 2026 – Present

Research Associate — Advanced Materials & Characterization

Sep 2024 – Sep 2026

New York University Abu Dhabi (NYUAD), Abu Dhabi, UAE

- Characterized structure-property relationships, flexural rigidity, and failure mechanisms in advanced architected metallic and composite structures under static loading conditions
- Qualified printed composite structures against quasi-static compression, flexural, modal, and frequency-response test protocols to validate strength, energy absorption, and vibration performance.
- Operated and maintained 10+ industrial additive manufacturing systems, including Raise3D Pro3 Plus HS, Stratasys J850/F370, Formlabs Form3, and Bambu Lab Carbon X1, supporting continuous production and prototyping throughput.

#### Postdoctoral Research Fellow — Metal 3D Printing & Smart Materials

Feb 2020 – Jun 2024

Khalifa University, Abu Dhabi, UAE

- Fabricated and qualified NiTi shape memory alloy TPMS lattice structures via Laser Powder Bed Fusion (EOS M400); optimized laser power, scan speed, and energy density to control microstructure and phase transformation behavior.
- Executed full materials characterization pipeline on 50+ printed samples using JEOL 7610F SEM, FEI Quanta 3D FIB-SEM (SEI/BSE/EDS), Bruker D2 and Rigaku MiniFlex XRD, Mettler Toledo and Setaram DSC, and Leica optical microscopy.

- Ran static (Instron 5969, 50 kN), fatigue (Instron 8872, 25 kN), and compression (MTS 100 kN) qualification campaigns; developed defect and print-quality maps used to reduce porosity and improve part consistency in AM NiTi components.
- Authored 4 peer-reviewed publications and supported 450+ citations of program-relevant materials data (Materials & Design, Journal of Manufacturing Processes, Materials, JMEP); reviewed 30+ technical manuscripts across 15 journals.
- Completed comprehensive EHS lab safety training covering chemical handling, HF acid etching, cryogenic substances, risk assessment, and emergency response protocols.

### Project Fellow — Smart & Functional Materials

Jun 2013 – Aug 2014

CSIR – Advanced Materials & Processes Research Institute (AMPRI), Bhopal, India

- Synthesized Cu-Al-Mn and Cu-Al-Ni shape memory alloys via vacuum induction melting under a Government of India (12th Five Year Plan) funded program; performed hot/cold rolling, heat treatment, quenching, and metallographic preparation.
- Characterized phase transformation and mechanical properties using SEM, XRD, DSC, and hardness/tensile testing; outputs contributed to 5 peer-reviewed publications and a research collaboration with IIT Madras.

### Engineering Intern — Steel Plant Operations

Jun – Jul 2011

Bokaro Steel Plant, Steel Authority of India Limited (SAIL), Jharkhand, India

- Gained direct industrial exposure to large-scale integrated steel manufacturing processes, plant operations, and quality systems.

## EDUCATION

### PhD, Materials Science & Technology

2014 – 2019

Academy of Scientific and Innovative Research (AcSIR) / CSIR-AMPRI, Bhopal, India — Thesis: Cu-based shape memory alloys: composition, processing, microstructure, and shape memory properties

### BEng, Mechanical Engineering

2008 – 2012

UIT, Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal, India

## TECHNICAL SKILLS

**Additive Manufacturing Systems:** EOS M400 (LPBF/SLM); Stratasys F370/J850 (FDM/PolyJet); Raise3D Pro3 Plus HS, Forge1 (FFF); Formlabs Form3 (SLA); Ultimaker S1

**Materials Characterization:** JEOL 7610F SEM; FEI Nova NanoSEM; FEI Quanta 3D FIB-SEM (SEI/BSE/EDS); Bruker D2 & Rigaku MiniFlex XRD; Mettler Toledo DSC1; Setaram DSC; Leica Optical Microscopy; Bruker Q4 Tasman

**Mechanical Testing & Processing:** Instron 5969 (static, 50 kN); Instron 8872 (fatigue, 25 kN); MTS 100 kN UTM; Controls Simplex 350 & Automax; vacuum induction melting; two-roll rolling; heat treatment; metallography; etching

**Design, Simulation & Computing:** SolidWorks; FreeCAD; Materialise Magics; GrabCAD Print; Bambu Studio; PreForm; IdeaMaker; Cura; Python (Pandas, NumPy, Scikit-learn); MATLAB; SQL; Linux; Shell scripting; LaTeX

## CERTIFICATIONS

- Additive Manufacturing Specialization — Arizona State University
- IBM Data Science Professional Certificate — IBM
- Responsible Conduct of Research — CITI Program, Florida
- Google AI Essentials Certificate — Google
- Python for Everybody Specialization — University of Michigan
- Lab Safety (EHS) — Khalifa University

## SELECTED PUBLICATIONS

500+ citations | h-index: 10 | ORCID: 0000-0002-4355-2169 | Peer reviewer for 15 journals including Additive Manufacturing, Materials & Design, Scripta Materialia

- Hussain S, Mehraj H, Karathanasopoulos N, Moustafa MA. Do Chopped Fibers Matter? Process-Structure-Property Relationships and Thermal Annealing Effects in Fiber-Reinforced TPMS Lattices. Progress in Additive Manufacturing, 2026.
- Hussain S, Alagha AN, Zaki W. Phase Transformation Behavior of NiTi TPMS Lattices Fabricated by LPBF. Journal of Materials Engineering & Performance, 2024.
- Hussain S, Alagha AN, Haidemenopoulos GN, Zaki W. Microstructural & Surface Analysis of NiTi TPMS Lattice Sections Fabricated by LPBF. Journal of Manufacturing Processes, 2023.

## PROFESSIONAL AFFILIATIONS

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- Technical Volunteer, Materials Research Society (MRS)
- Individual Member, Research Data Alliance (RDA)
- Associate Member, European Materials Modelling Council (EMMC)
- Member, AM Platform: European Technology Platform in Additive Manufacturing

## ADDITIONAL INFORMATION

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**Languages:** English (Professional), Hindi (Native), Urdu (Native)

**Profiles:** [github.com/drshahadathussain](https://github.com/drshahadathussain) | [researchgate.net/profile/Shahadat-Hussain](https://researchgate.net/profile/Shahadat-Hussain) |

[scholar.google.com/citations?user=tQNSWaAAAAAJ](https://scholar.google.com/citations?user=tQNSWaAAAAAJ)